Smart Steel Factory Automation

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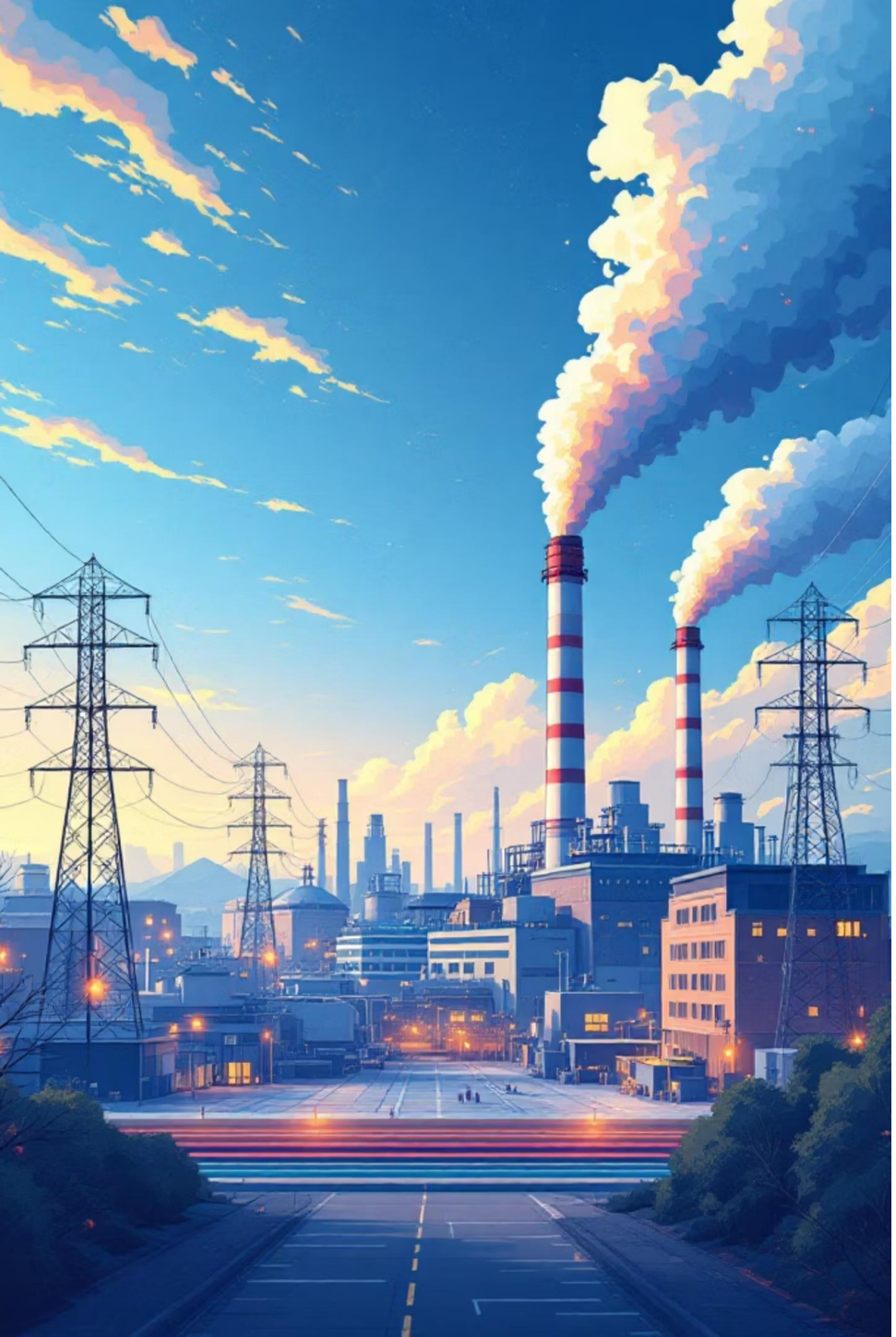


Introduction

Following the SDGs and With climate change and increasing the global warming, and after all the current power crisis because of closing Strait of Hormuz, factories must become smart and automated to compete, especially steel plants as they are using induction furnaces that draw very high reactive power and create lagging current.

Using AI, this report addresses big data not only to reduce costs, but also to optimize resources, connect manufacturing with a healthier environment, and prove that technology can reduce costs without reducing quality.

Business Problem Analysis

- Big Lagging reactive current
- Huge CO_2 emissions
- High penalties and bills
- Low Power factor: 0.65
- Poor efficiency and thermal stress
- No clear data for decision-making

Penalties System according to EgyptERA

According to EgyptERA, penalties start when power factor falls below 0.92. In this case, the factory's PF is 0.65, which means high penalties must be paid to the government.

THE CURRENT TARIFF IS APPLIED BASED ON THE FOLLOWING:-

- PRICES ARE APPLIED BASED ON A POWER FACTOR OF 0.92
- PRICES ARE DETERMINED ACCORDING TO THE EXCHANGE RATE ANNOUNCED BY THE CENTRAL BANK (30.9 EGP/USD),WITH THE TARIFF TO BE RECONSIDEREDIN IN CASE OF AN EXCHANGE RATE CHANGE.
- CAPACITY CHARGES ARE APPLIED BASED ON THE MAXIMUM USER LOAD EVERY THREE MONTHS.
- TIME-OF-USE TARIFFS ARE APPLIED ACCORDING TO THE SMART METER APPLICATION PROGRAM, WITH PEAK HOURS DETERMINED BY THE MINISTRY OF ELECTRICITY AND ENERGY LASTING 4 HOURS.
- IN THE ABSENCE OF METERS, THE AVERAGE ENERGY PRICE IS APPLIED

Alignment with Sustainable Development Goals

SDG 7 — Affordable & Clean Energy

Predictive PF control reduces energy waste and supports national energy efficiency targets.

SDG 9 — Industry & Innovation

Digital twin and AI replace reactive maintenance with proactive, resilient infrastructure.

SDG 12 — Responsible Production

Computer vision quality control reduces scrap and material waste on the line.

SDG 13 — Climate Action

Lower reactive power and optimized energy use directly reduce CO2 emissions.

Raw IoT Data Stream

The data type is structured and stochastic consisting of discrete readings by IoT sensors for each 15 mins through 1 year

	A	B	C	D	E	F	G	H	I	J	K	L
1	date	Usage_kWh	Lagging_Current_Reactive_Power_kVarh	Leading_Current_Reactive_Power_kVarh	CO2(tCO2)	Lagging_Current_Power_Factor	Leading_Current_Power_Factor	NSM	WeekStatus	Day_of_week	Load_Type	Power Fac
14427	31/05/2018 06:30	2.84	5.36	0	0	46.82	100	23400	Weekday	Thursday	Light_Load	0.47
14428	31/05/2018 06:45	2.88	5.33	0	0	47.54	100	24300	Weekday	Thursday	Light_Load	0.48
14429	31/05/2018 07:00	2.92	5.29	0	0	48.33	100	25200	Weekday	Thursday	Light_Load	0.48
14430	31/05/2018 07:15	2.92	5.26	0	0	48.54	100	26100	Weekday	Thursday	Light_Load	0.49
14431	31/05/2018 07:30	3.67	5.15	0	0	58.03	100	27000	Weekday	Thursday	Light_Load	0.58
14432	31/05/2018 07:45	4.25	6.26	0	0	56.17	100	27900	Weekday	Thursday	Light_Load	0.56
14433	31/05/2018 08:00	4.61	7.2	0	0	53.92	100	28800	Weekday	Thursday	Light_Load	0.54
14434	31/05/2018 08:15	66.46	47.74	0	0.03	81.22	100	29700	Weekday	Thursday	Light_Load	0.81
14435	31/05/2018 08:30	80.32	63.4	0	0.04	78.49	100	30600	Weekday	Thursday	Light_Load	0.78
14436	31/05/2018 08:45	67.75	64.15	0	0.03	72.61	100	31500	Weekday	Thursday	Light_Load	0.73
14437	31/05/2018 09:00	133.13	71.78	0	0.06	88.02	100	32400	Weekday	Thursday	Light_Load	0.88
14438	31/05/2018 09:15	95.22	38.7	0	0.04	92.64	100	33300	Weekday	Thursday	Medium_Load	0.93
14439	31/05/2018 09:30	54.54	27.14	0	0.03	89.53	100	34200	Weekday	Thursday	Medium_Load	0.9
14440	31/05/2018 09:45	115.16	47.02	0	0.05	92.58	100	35100	Weekday	Thursday	Medium_Load	0.93
14441	31/05/2018 10:00	121.46	63.72	0	0.06	88.55	100	36000	Weekday	Thursday	Medium_Load	0.89
14442	31/05/2018 10:15	99.68	38.45	0	0.05	93.3	100	36900	Weekday	Thursday	Maximum_Load	0.93
14443	31/05/2018 10:30	110.09	43.99	0	0.05	92.86	100	37800	Weekday	Thursday	Maximum_Load	0.93
14444	31/05/2018 10:45	49.75	24.55	0	0.02	89.68	100	38700	Weekday	Thursday	Maximum_Load	0.9
14445	31/05/2018 11:00	57.89	32.36	0	0.03	87.29	100	39600	Weekday	Thursday	Maximum_Load	0.87
14446	31/05/2018 11:15	60.3	44.28	0	0.03	80.6	100	40500	Weekday	Thursday	Maximum_Load	0.81
14447	31/05/2018 11:30	46.01	36.25	0	0.02	78.55	100	41400	Weekday	Thursday	Maximum_Load	0.79
14448	31/05/2018 11:45	64.04	56.95	0	0.03	74.73	100	42300	Weekday	Thursday	Maximum_Load	0.75
14449	31/05/2018 12:00	47.16	34.42	0	0.02	80.77	100	43200	Weekday	Thursday	Maximum_Load	0.81
14450	31/05/2018 12:15	6.59	0.54	18.68	0	99.67	33.27	44100	Weekday	Thursday	Medium_Load	0.33
14451	31/05/2018 12:30	5.08	0	25.2	0	100	19.76	45000	Weekday	Thursday	Medium_Load	0.2

EDA, Cleaning, and Processing

01

Upload

Steel Industry CSV data was loaded into Google Colab to start preprocessing .

02

Cleaning (Preprocessing)

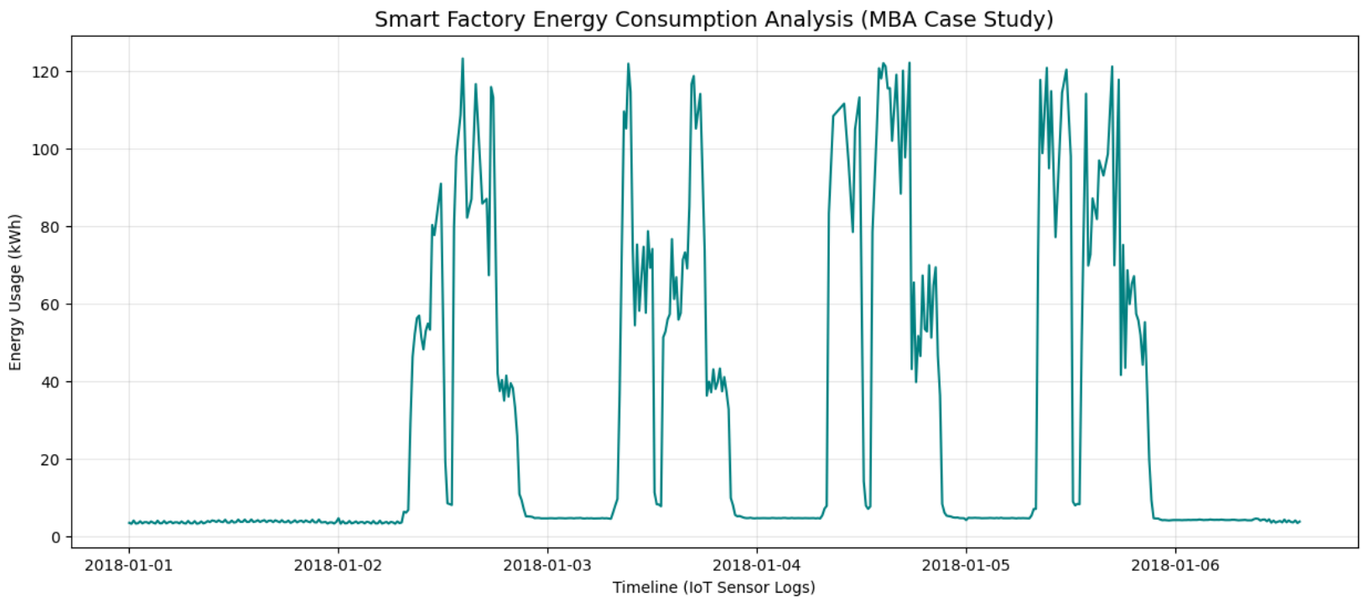
Nulls and outliers were removed.

03

Insights (processing)

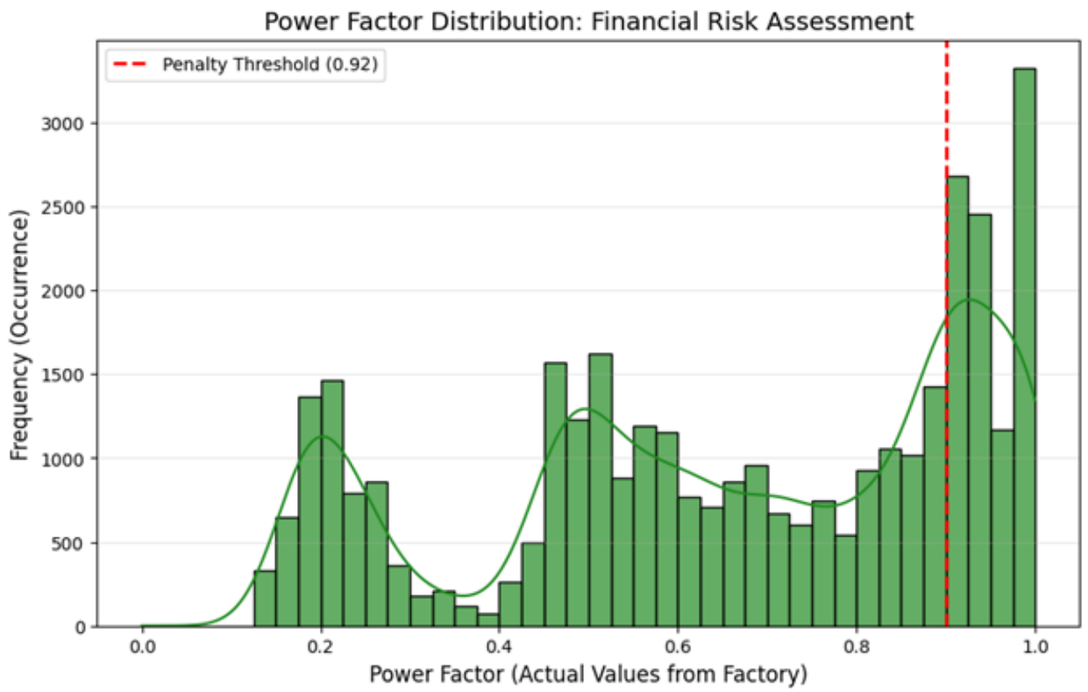
Checking all the insights by AI and taking the most important ones –Check next slides-

Key Data Insights



Graph 1

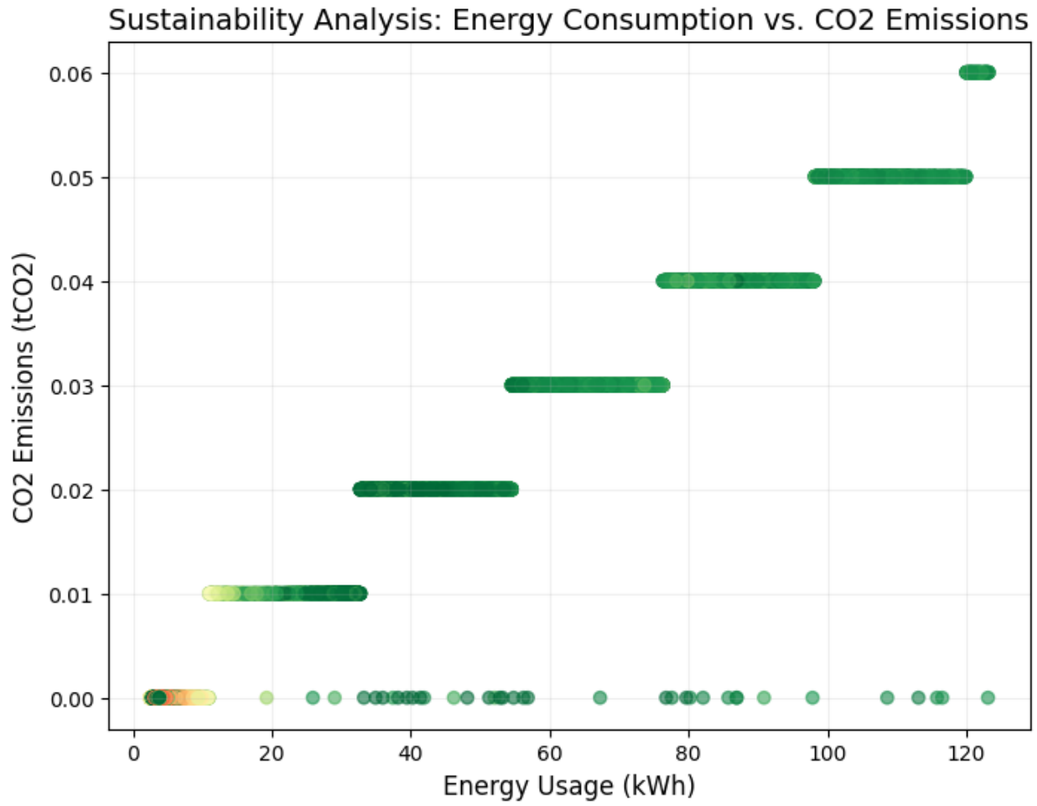
The power consumption followed normal furnace cycles on & off, confirming that there is no issue.



Graph 2

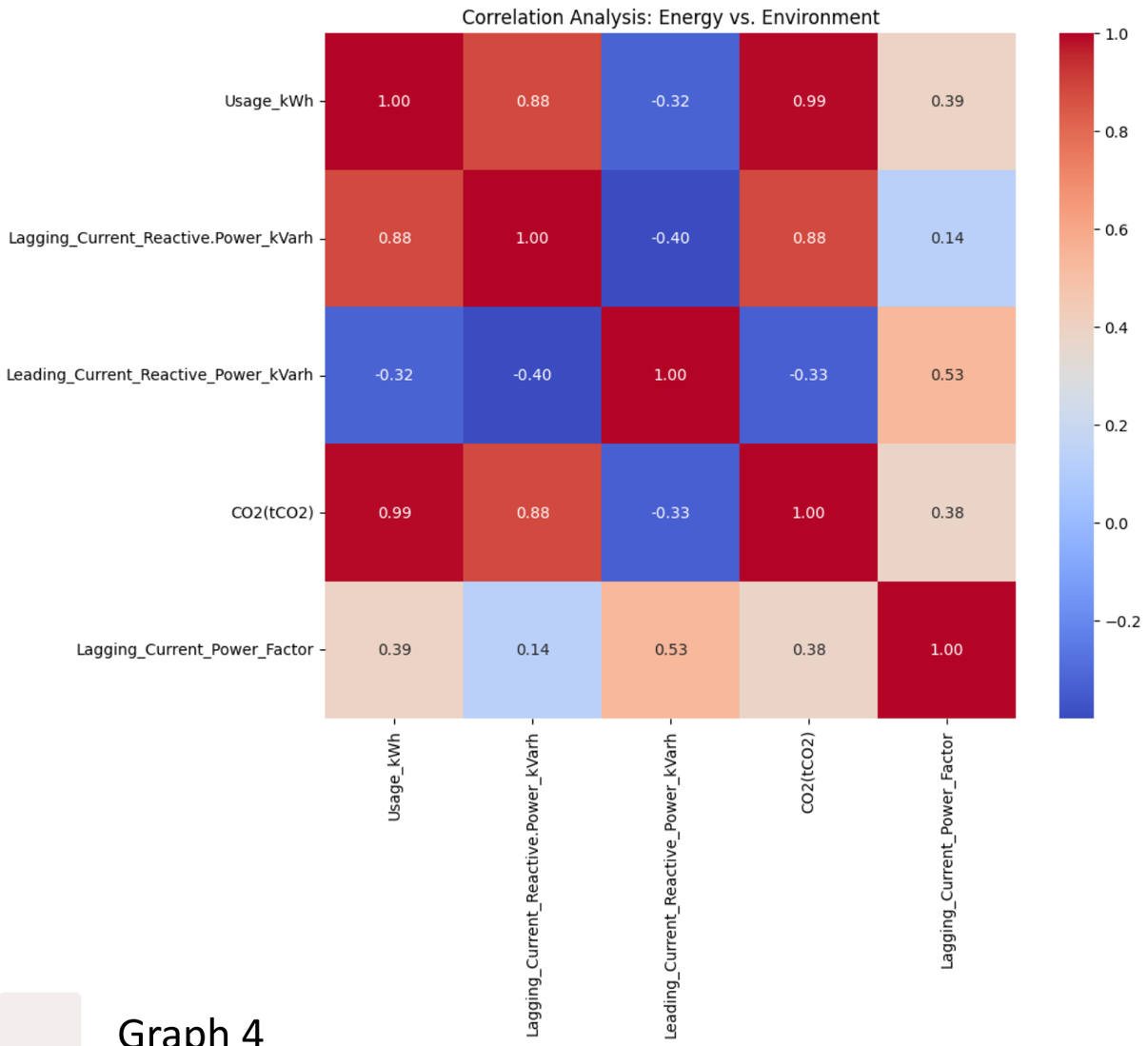
Most PF readings are below 0.92, with an average of about 0.65, so the factory pays huge penalties.

Key Data Insights



Graph 3

Energy consumption and CO2 emissions are directly proportional



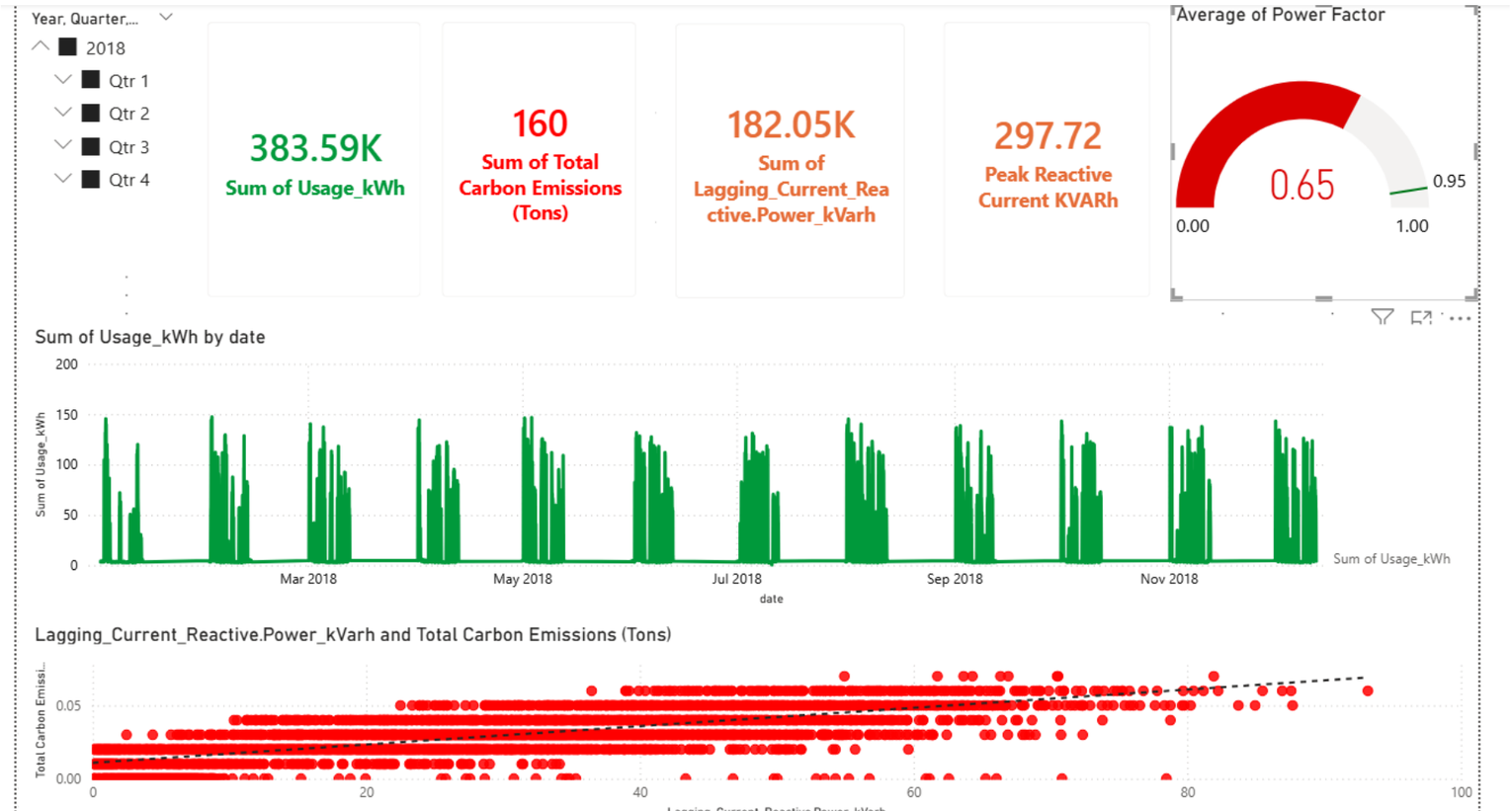
Graph 4

KWh and KVARh are directly proportional, CO2 and Kvarh are also directly proportional at 0.88, confirming the need to reduce lagging reactive power to reduce CO2 emissions and power consumption.

Power BI Dashboard for Top Management

The dashboard provides four key cards: total active power usage, total carbon emissions, total lagging reactive power, and peak lagging reactive power.

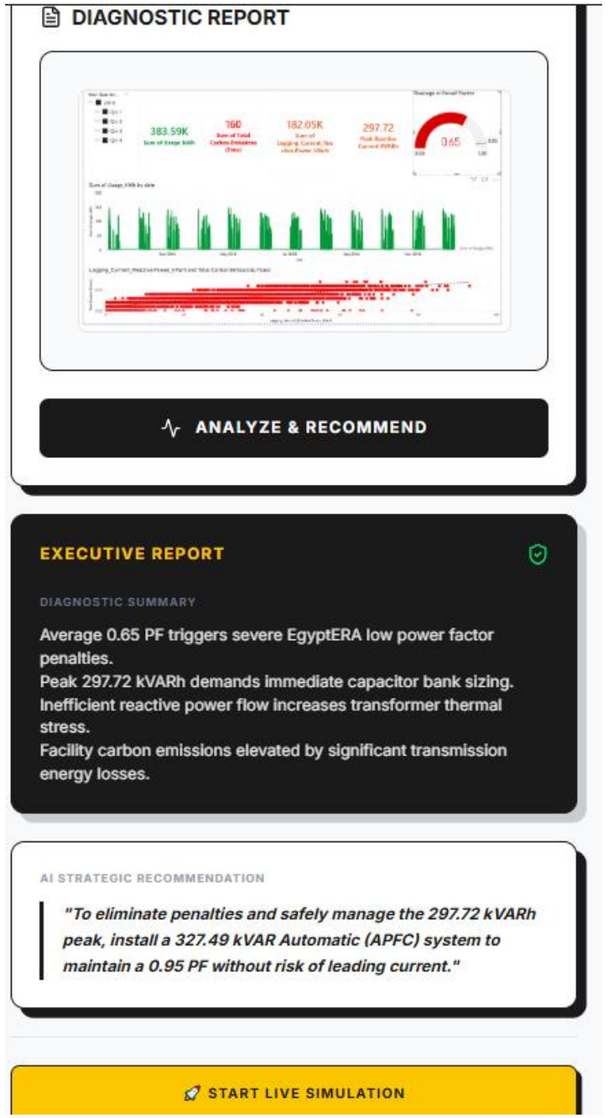
It also shows three graphs: a PF gauge, energy consumption over time, and lagging reactive power versus CO2 emissions. This supports data-driven decisions instead of experience alone.

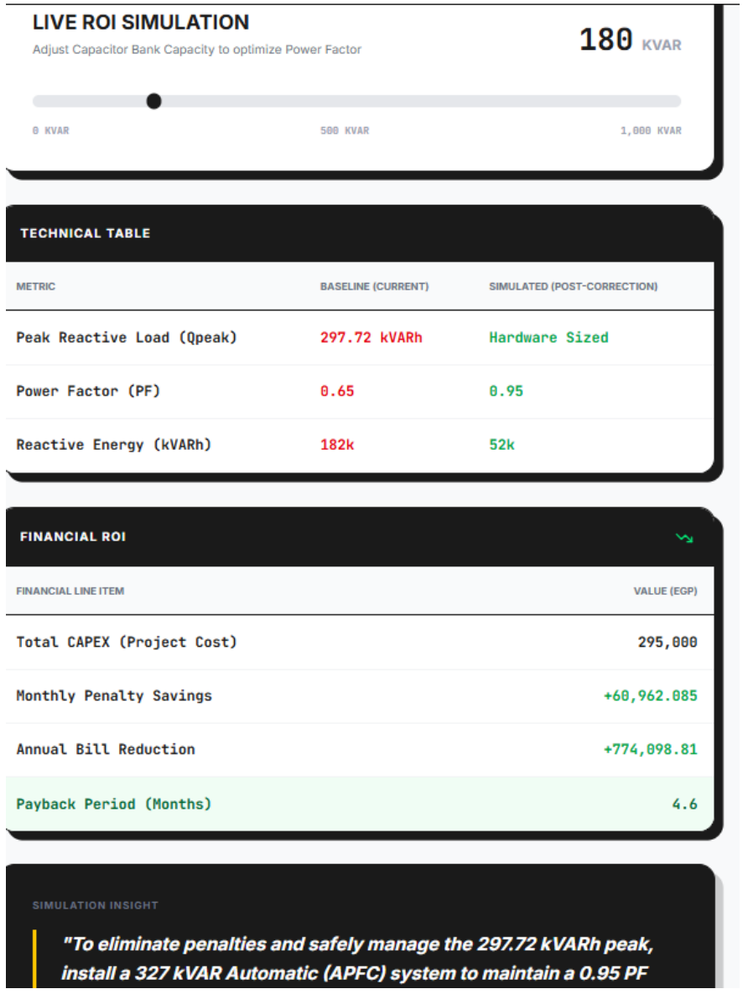


Solution Using AI Studio Application

The AI Studio application uses computer vision to read the Power BI dashboard from a screenshot, then analyzes the PF, lagging current, and CO2 emissions for the top management.

By starting live simulation, it tests capacitor values, searches the internet for costs and installation fees, and calculates penalty reduction and live ROI for the APFC investment.





AI Solutions — Applications & Algorithms

Random Forest Regressor

Predicts PF for next 15-min window using Usage_kW, Lagging/Leading currents and engineered features.

Computer Vision (CNN)

Automated quality inspection to reduce scrap and support SDG 12.

Reinforcement Learning

Energy-aware scheduling to shift heavy tasks off-peak; research suggests up to 22% savings.

The SteelTwin AI application (digital twin) performs data acquisition → inference → prescriptive action. If predicted PF < 0.92, the system recommends/automates capacitor bank engagement; it also simulates capacitor sizes, fetches (assumed) costs, estimates penalty and bill reductions, and computes ROI.

ERP Integration & Operationalizing

Automated Work Orders

AI action flags generate maintenance tasks and check spares inventory.

Financial & ESG Reporting

kgCO2 and energy savings update ERP financials and sustainability ledgers in real time.

Value Adds

Data-driven procurement, inventory optimization (JIT), and ESG compliance.

Strategic Recommendations & Limitations

Recommendations

- Automate control loop: AI → PLC to engage capacitors when PF prediction < 0.92.
- Integrate AI via API to ERP for automated maintenance and ESG reporting.
- Expand model for energy-aware scheduling to capture up to 22% cost savings.
- Scale the digital twin across lines and enforce data governance & dynamic re-training.

Limitations

Data quality, interoperability, cybersecurity, data privacy, model drift, and high initial costs can impede deployment. Ongoing re-training and robust security are required.

References: Berawi (2019); Nozari et al. (2025); Rehan (2021); Shaikat (2026).

Executive Summary

The Problem

Heavy induction furnaces operate at a **0.65 Power Factor**—well below EgyptERA's **0.92 penalty threshold**. This **inefficiency** triggers massive **government fines** and releases **404 tons of CO_2**.

The Intervention

A **proprietary AI pipeline** analyzed **35,000 IoT data points** to simulate a **Digital Twin environment**, calculating the precise **Automatic Power Factor Correction (APFC)** needed.

The Outcome

Upgrading the factory to a **0.95 Power Factor**, **100% eliminating government penalties**, **reducing thermal cable stress**, and drastically **lowering the carbon footprint** with a **verified ROI**.